

```

# =====
# PYTHON QUICK REVISION FILE (5-10 Minutes)
# Topics: print, comments, variables, type(), data types, input,
#         type casting, small demo
# =====

# -----
# PRINT FUNCTION
# -----

# print() shows output on screen
# Used to display values, messages, results

print("Hello Python")

# Python executes line by line

# -----
# COMMENTS
# -----

# Single line comment starts with #

"""
Multiline comments are used for longer explanation.
Python ignores these lines completely.
Comments are for humans, not for Python.
"""

# -----
# VARIABLES
# -----

# Variable = container to store data

name = "Rahul"
age = 21
marks = 85.5

print(name)
print(age)
print(marks)

# Python decides datatype automatically
# This is called DYNAMIC TYPING

# -----
# type() FUNCTION
# -----

# type() tells us datatype of variable

print(type(name))
print(type(age))
print(type(marks))

# -----
# BASIC DATA TYPES
# -----

# int    → whole numbers
# float  → decimal numbers
# str    → text
# bool   → True / False

students = 40      # int
percentage = 78.5  # float
city = "Delhi"     # string
is_present = True  # boolean

```

```

print(type(students))
print(type(percentage))
print(type(city))
print(type(is_present))

# -----
# WHY DATA TYPES MATTER
# -----

print(10 + 5)      # numeric addition
print("10" + "5")  # string joining

# Same values.
# Different datatype.
# Different result.

# -----
# USER INPUT
# -----

# input() always gives STRING

user_age = input("Enter age: ")
a = input()
print(type(user_age))

# -----
# TYPE CASTING
# -----

# Converting one datatype into another

age = int(user_age)      # string → int
salary = float("45000.5") # string → float
number = str(100)        # int → string

print(age)
print(salary)
print(number)

# -----
# REAL LIFE MINI DEMO
# -----

m1 = "70.5"
m2 = "80.0"
m3 = "90.5"

f1 = float(m1)
f2 = float(m2)
f3 = float(m3)

age = int(input("Age :"))
print(age + 5)

average = (f1 + f2 + f3) / 3

std_code = int(input("enter std :"))
phone_number = int(input("Enter Phone Number : "))

# std_code = input("enter std :")
# phone_number = input("Enter Phone Number : ")

# print("std_code" + "phone_number")

# print("std_code" + "phone_number")

print(std_code + phone_number)

print("Average Marks:", average)

message_1 = input("Enter Message :")

```

```

# -----
# FINAL REVISION SUMMARY
# -----

# ✓ print()
# ✓ comments
# ✓ variables
# ✓ type()
# ✓ int, float, str, bool
# ✓ input()
# ✓ type casting
# ✓ basic analysis

print("Revision Complete Ready for next topic!")

```

```

Hello Python
Rahul
21
85.5
<class 'str'>
<class 'int'>
<class 'float'>
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
15
105
Enter age: 23
bdd
<class 'str'>
23
45000.5
100
Age :90
95
enter std :9
Enter Phone Number : 9
18
Average Marks: 80.33333333333333
Enter Message :90
Revision Complete Ready for next topic!

```

```

# =====
# PYTHON SESSION 2 – STRINGS
# Topics:
# - What is String
# - Why Strings are Important
# - Indexing
# - Slicing
# - String Methods (upper, lower, replace, split)
# - String Formatting
# - Real-life Cleaning Demo
# =====

# -----
# 1. WHAT IS A STRING?
# -----

# String = collection of characters
# Used to store TEXT data.

# Real life examples:
# - Names
# - Cities
# - Product names
# - Emails
# - Addresses

name = "Sahil"
product = "Laptop"

print(name)
print(product)

```

```

# -----
# WHY STRINGS EXIST
# -----

# Computers deal with TWO things:
# - Numbers
# - Text

# Numbers are used for calculation.
# Text is used for information.

# You cannot calculate with names or cities.
# That's why Python created STRING datatype.

# Example:
# You can ADD numbers
# You CANNOT add names mathematically.

# -----
# 2. STRING INDEXING
# -----

# Every character in string has position (index).
# Index starts from 0.

city = "Mumbai"

# M u m b a i   c i t y
# 0 1 2 3 4 5 6 7 8 9 10

print(city[0])  # M
print(city[1])  # u
print(city[-1]) # i (last character)

# Why indexing is useful:
# To access specific characters.

# -----
# 3. STRING SLICING
# -----

# Slicing = extracting part of string

text = "PythonProgramming"

print(text[0:6])    # Python
print(text[6:])     # Programming
print(text[:6])     # Python

# Syntax:
# string[start:end]

# -----
# 4. STRING METHODS
# -----

# Methods are built in functions for strings.

# -----
# upper()
# -----

course = "python"

print(course.upper())

# Converts all characters to uppercase

# -----
# lower()
# -----

```

```

brand = "SAMSUNG"

print(brand.lower())

# Converts all characters to lowercase

# -----
# capitalize()
# -----

brand = "SAMSUNG"

print(brand.capitalize())

# Converts all characters to "first letter capital rest small ""

# -----
# replace()
# -----

product_name = "Apple Phone"

print(product_name.replace("Phone", "Mobile"))

# Replaces one word with another

# -----
# split()
# -----

sentence = "Python is very easy"

words = sentence.split(" ")

print(words)

# Splits string into list based on separator

# -----
# 5. STRING FORMATTING
# -----

# Used to combine variables with text

name = "Rahul"
age = 21

print("My name is", name, "and age is", age)

# Better formatting:

print(f"My name is {name} and age is {age}")

# f-string is modern and clean.

# -----
# 6. REAL LIFE DEMO – CLEAN PRODUCT NAME
# -----

# Raw product data (messy input)

raw_product = " samsung-Mobile "

print("Original:", raw_product)

# Step 1: remove spaces
clean1 = raw_product.strip()

# Step 2: convert to lowercase
clean2 = clean1.lower()

# Step 3: replace dash with space
final_product = clean2.replace("-", " ")

```

```

print("Cleaned Product:", final_product)

# Final output:
# samsung mobile

# This is REAL data cleaning.
# Used in Data Science and ML pipelines.

# -----
# WHY STRING CLEANING IS IMPORTANT
# -----

# Because real-world data is:
# - messy
# - inconsistent
# - contains extra spaces
# - mixed case

# Cleaning ensures:
# - consistency
# - accuracy
# - better analysis

# -----
# MINI PRACTICE FOR STUDENTS
# -----

student_name = input("Enter your name: ")

print(student_name.upper())
print(student_name.lower())
print(student_name[0:3])

# -----
# SESSION 2 SUMMARY
# -----

# ✓ What is string
# ✓ Why strings exist
# ✓ Indexing
# ✓ Slicing
# ✓ upper()
# ✓ lower()
# ✓ replace()
# ✓ split()
# ✓ formatting
# ✓ product cleaning demo

print("SESSION 2 COMPLETE – STRINGS MASTERED ")

```

```

Sahil
Laptop
M
u
i
Python
Programming
Python
PYTHON
samsung
Samsung
Apple Mobile
['Python', 'is', 'very', 'easy']
My name is and age is 21
My name is Rahul and age is 21
Original:  samsung-Mobile
Cleaned Product: samsung mobile
Enter your name: 45
45
45
45
SESSION 2 COMPLETE – STRINGS MASTERED

```

```
# std_code = int(input("enter std :"))
# phone_number = int(input("Enter Phone Number : "))

std_code = input("enter std :")
phone_number = input("Enter Phone Number : ")
print(type(std_code))
print(type(phone_number))

# print("std_code" + "phone_number")

# print("std_code" + "phone_number")

print(std_code + phone_number)
```

```
enter std :91
Enter Phone Number : 1234567899
<class 'str'>
<class 'str'>
911234567899
```

you have to take input from user as string that should be his/her complete addresses then you have to slice it according to

1. house/flat no.
2. street name
3. city
4. state
5. Country
6. Pin code

▼ =====

PYTHON REVISION QUESTIONS (SESSION 1 + SESSION 2)

=====

Q1. Print "Welcome to Python" on screen.

Q2. Create a variable called city and store your city name.

Print its datatype.

Q3. Store your age in a variable and check its type.

Q4. Take user input for marks and print its datatype.

Why is it string?

Q5. Convert the marks input into float and print new datatype.

Q6. What is the difference between:

```
print(10 + 5)
```

```
print("10" + "5")
```

Q7. Create a string "Programming" and print:

a) First character

b) Last character

Q8. From "PythonProgramming", extract only "Python" using slicing.

Q9. Convert "data science" into uppercase.

Q10. Convert "MACHINE LEARNING" into lowercase.

Q11. Replace "Phone" with "Mobile" in "Apple Phone".

Q12. Split this sentence into words:

"Python is very powerful"

Q13. Use f-string to print:

My name is Rahul and age is 21

Q14. Clean this product name:

" samsung-Mobile "

Final output should be:

samsung mobile

Q15. Write one line explaining why data types are important.

Start coding or [generate](#) with AI.

✓ =====

PYTHON REVISION QUESTIONS — SESSION 1 + SESSION 2

=====

Q1. Print "Learning Python is fun!" on screen.

Q2. Create a variable called country and store "India".

Print its datatype using type().

Q3. Store your birth year in a variable and print its datatype.

Q4. Take user input for salary and print its datatype.

Why does input() always return string?

Q5. Convert the salary input into float and print the new datatype.

Q6. Observe and explain:

```
print(20 + 10)
```

```
print("20" + "10")
```

Q7. Create a string "DataAnalytics" and print:

a) First character

b) Last character

Q8. From "MachineLearning", extract only "Machine" using slicing.

Q9. Convert "artificial intelligence" into uppercase.

Q10. Convert "BIG DATA" into lowercase.

Q11. Replace "Old" with "New" in:

"Old Laptop"

Q12. Split this sentence into words:

"Python makes coding easy"

Q13. Use f-string to print:

My city is Mumbai and my age is 22

Q14. Real life cleaning:

Raw product name:

" HP-Laptop "

Perform these steps:

- remove spaces
- convert to lowercase
- replace "-" with space

Final output should be:

hp laptop

Q15. Hardest Question:

Take input for product price as string,

convert it to float,

add 18% GST,

and print final price.

Example:

Input: "1000"

Output: 1180.0

Write one line explaining WHY type casting is needed.

```
price=input("enter price")
price=float(price)
final_price=(price+(price*0.18))
print(final_price)
```

```
enter price200
236.0
```

Start coding or [generate](#) with AI.